🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.AT.8

Solve and Graph Inequalities

Lesson 7-6 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can solve an inequality and graph its solution set on a number line.
Language Objective: I can explain my work using the words solve, graph, solution, and substitute.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Solve | Graph | Solution | Substitute | Solution set | Inverse operation

👆 *Tap any word to see what it means and a picture.*

1. To find all the values that make an inequality true is to ______ it.

2. To show the solutions on a number line is to ______ them.

3. A value that makes an inequality true is a ______.

4. To check a solution, I ______ the value back into the inequality.

5. All the values that make an inequality true form the ______.

6. An operation that undoes another to isolate the variable is an ______.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

What steps did you follow to solve and then graph the inequality, and how did you check a value to make sure your solution set is right?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Solve** — To find the number that makes it true.
  Resolver — Encontrar el número que la hace verdadera.
- [ ] **Graph** — To show answers on a number line with circles and shading.
  Graficar — Mostrar respuestas en una recta numérica con círculos y sombreado.
- [ ] **Solution** — A number that makes the equation or inequality true.
  Solución — Un número que hace verdadera la ecuación o desigualdad.
- [ ] **Substitute** — To put a number in for the letter to check if it works.
  Sustituir — Poner un número en lugar de la letra para ver si funciona.
- [ ] **Solution set** — All of the values that make an inequality true.
  Conjunto solución — Todos los valores que hacen verdadera una desigualdad.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

First I solved by ____, then I graphed ____.
Primero resolví al ____, luego grafiqué ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Solving an inequality gives a range of solutions, not one number.

Evidence: Students note both use inverse operations to isolate the variable, but an inequality's solution is many values (a range) while an equation's is a single value.

Reasoning: This evidence connects the definition of solve to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **First I solved by ____, then I graphed ____.**
  Primero resolví al ____, luego grafiqué ____.
- **I substituted ____ to check that it ____ the inequality.**
  Sustituí ____ para verificar que ____ la desigualdad.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Solve
2. **Notes 2:** Graph
3. **Notes 3:** Solution
4. **Notes 4:** Substitute
5. **Notes 5:** Solution set
6. **Notes 6:** Inverse operation
7. **Watch for this mistake:** *A common mistake in Solve and Graph Inequalities is shading the number line in the wrong direction after solving. For example, a student solves $x + 4 > 10$ correctly to get $x > 6$, but then shades LEFT toward smaller numbers instead of RIGHT toward greater numbers. The shading direction always matches the solved symbol, not the original problem: '>' and '≥' shade right (greater numbers), '<' and '≤' shade left (smaller numbers) — check this AFTER you solve, not before. To catch this mistake, substitute one shaded value and one unshaded value back into the original inequality: the shaded value should make it true, and the unshaded value should make it false.*
8. **Try It 1:** A. Yes, because $5 \leq 5$ is true — *Substitute $x = 5$ into $x \leq 5$: $5 \leq 5$ is true because ≤ means 'less than OR equal to.' The boundary value 5 IS a solution, so the gate opens.*
9. **Try It 2:** A. All numbers greater than 7, like 8, 9, 10, ... (but NOT 7 itself) — *The solution set is every value that makes $x > 7$ true. That means all numbers greater than 7, such as 8, 9, 10, and beyond. Because > does not include the boundary, 7 itself is NOT in the solution set.*